Ian Z. Branigan

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Research Interests: Exoplanet and brown dwarf characterization via spectroscopy. Disequilibrium chemistry, atmospheric dynamics, cloud and haze physics, interior-atmosphere interactions, and terrestrial surfaces; and their connections to observed spectra, including time variability.

Education

Cornell University, College of Arts and Sciences, Ithaca, NY — Aug 2023–May 2027 (projected)

B.A. in Physics (Inside Concentration) | Minor in Astronomy

Major GPA: 4.11/4.3 [4.0/4.0] | Overall GPA: 4.09/4.3 [3.99/4.0]

Senior Thesis (in progress): *Revealing the atmospheric chemistry of a diverse range of exoplanets*

Selected Coursework: Planetary Physics, Planetary Atmospheres, Astrophysical Processes, Multiwavelength Astronomical Techniques, Mathematical Physics, Computational Engineering Physics, Applications of Quantum Mechanics, Analytical Mechanics, Advanced Electricity and Magnetism, Statistical Thermodynamics, Teaching and Learning Physics, Numerical Analysis

North Carolina School of Science and Mathematics, Durham, NC — Aug 2021–May 2023

D.H. Conley High School, Greenville, NC — Aug 2019–May 2021

Honors and Awards

Cranson and Edna B. Shelley Prize for Undergraduate Research, awarded for "excellence in research in astronomy." ***Cornell Dept of Astronomy, May 2026.***

Japanese Language Program Robert Sukle Award, awarded for "three years of hard work, dedication, and love of learning the Japanese language." *Cornell Dept of Asian Studies, May 2026.*

Roger and Mary Lou West Undergraduate Research Fellowship, $7000 awarded to support my research in disequilibrium chemistry in the atmosphere of WASP-39b conducted during summer 2025. ***Cornell Dept of Astronomy.***

James E. Rice Jr. Prize, awarded for first-year writing seminar essay in philosophy "Causation and Dōgen's Uji." *Cornell John S. Knight Institute for Writing in the Disciplines, Sep 2024.*

Elmer Markham Johnson Prize, awarded for first-year writing seminar essay in Asian history "Ethnicity and Imperialism in the Annexation of Ryūkyū." *Cornell John S. Knight Institute for Writing in the Disciplines, Apr 2024.*

Hunter R. Rawlings III Cornell Presidential Research Scholar, $8000 awarded to support my undergraduate research; awarded to <2% of undergraduates. ***Cornell University, Aug 2023–Present.***

Research Experience

Reanalysis of the NIRSpec PRISM Spectrum of a Sub-Neptune Exoplanet May 2026–Present

Chalmers University of Technology, Gothenburg, Sweden

Mentors: Prof. Jonathan Tan, Giuseppe Morello, Gareb Fernández Rodríguez

- Conducted a suite of free chemistry and equilibrium chemistry retrievals with POSEIDON to investigate the properties and possible formation pathways of LP 791-18c, a sub-Neptune with an apparently aerosol-rich atmosphere.
- Identified degeneracies between assumed aerosol optical properties and retrieved composition.
- Experimented with extending disequilibrium chemistry retrieval framework from previous work (below) to sub-Neptunes.

Retrievals for Probing Disequilibrium Processes in Gas Giant Exoplanets Jan 2024–Present

Cornell Department of Astronomy, Ithaca, NY

Mentors: Prof. Nikole Lewis, Elijah Mullens, Ryan Challener

- Developed and implemented a disequilibrium chemistry retrieval methodology in the open-source Python retrieval code POSEIDON.
- Parallelized the photochemical kinetics codes VULCAN and photochem in order to construct grids of 1D atmospheric disequilibrium chemistry models for the hot Jupiter exoplanet WASP-39b.
- Performed grid retrievals on the complete JWST transmission spectrum of WASP-39b using POSEIDON to place constraints on the atmospheric temperature profile, composition, and eddy diffusivity. Compared results to traditional free chemistry and equilibrium chemistry retrievals, which I also conducted.

Bayesian Inference of Exoplanet Transit Depths with Kepler and TESS Jan 2022–Dec 2022

North Carolina School of Science and Mathematics, Durham, NC

Mentor: Jonathan Bennett

- Conceptualized and carried out an independent project to compute Kepler/K2 and TESS transit depths of 24 exoplanets, using Python packages Lightkurve (handling time series data) and exoplanet (lightcurve fitting with MCMC).

Community Involvement and Leadership

Japanese Language Tutor Aug 2026–Present

Cornell Learning Strategies Center

- Holding weekly tutoring hours for first- and second-year students of Japanese, coaching students in effective strategies for language learning.

Physics Learning Assistant Jan 2026–May 2026

Cornell Department of Physics

- Led weekly study halls for students in PHYS 3318 (Analytical Mechanics), guiding students through learning upper-level physics concepts and helping them develop problem-solving skills.
- Attended weekly professional development course, becoming familiar with literature on pedagogy in physics and writing self-reflections on the efficacy of my teaching.

Co-President, Kendo Club at Cornell Jul 2025–May 2026

- Managed activities of the 7-member executive board of the club (e.g., coordination of tri-weekly practices, logistics of travel and lodging for tournaments, allocation of club finances).
- Instructed new club members in striking, etiquette, and responsibly maintaining equipment, with a focus on aspects critical to safe practice of the martial art.

Science Olympiad Volunteer Test Writing Aug 2023–Present

- Developing exam questions to encourage critical thinking related to fundamental astronomical concepts (principles of observational astronomy, stellar astrophysics, etc.) and recent research breakthroughs (exoplanet detection and characterization, missions to solar system objects, transient astrophysics, etc.) for 9 middle and high school competitions.

Publication and Presentations

I. Z. Branigan, E. Mullens, R. C. Challener, N. K. Lewis (2026) "A Novel Retrieval Technique to Probe Photochemical and Mixing Processes in Exoplanets with VULCAN," *In Prep*

Asia Oceania Geosciences Society (AOGS) Meeting. Fukuoka, Japan. (**Oral**)	Aug 2026
CICO Science Workshop: Planet Formation & Exoplanet Evolution—from JWST to Ariel (**Oral**)	Jun 2026
Winter American Astronomical Society (AAS) Meeting. Phoenix, AZ. (**Oral**)	Jan 2026
Cornell Astronomy Undergraduate Research Poster Forum. Ithaca, NY. (**Poster**)	May 2025
NC Section of the American Association of Physics Teachers Meeting. Greensboro, NC. (**Poster**)	Nov 2022
North Carolina Astronomers' Meeting. Virtual. (**Oral**)	Sep 2022

Technical Skills

Programming languages: Python (NumPy, Matplotlib, SciPy, Numba), Julia, Java
Supplementary tools: Unix/Zsh, Git/GitHub, Slurm, LaTeX
Laboratory experience: CCDs, SDRs
